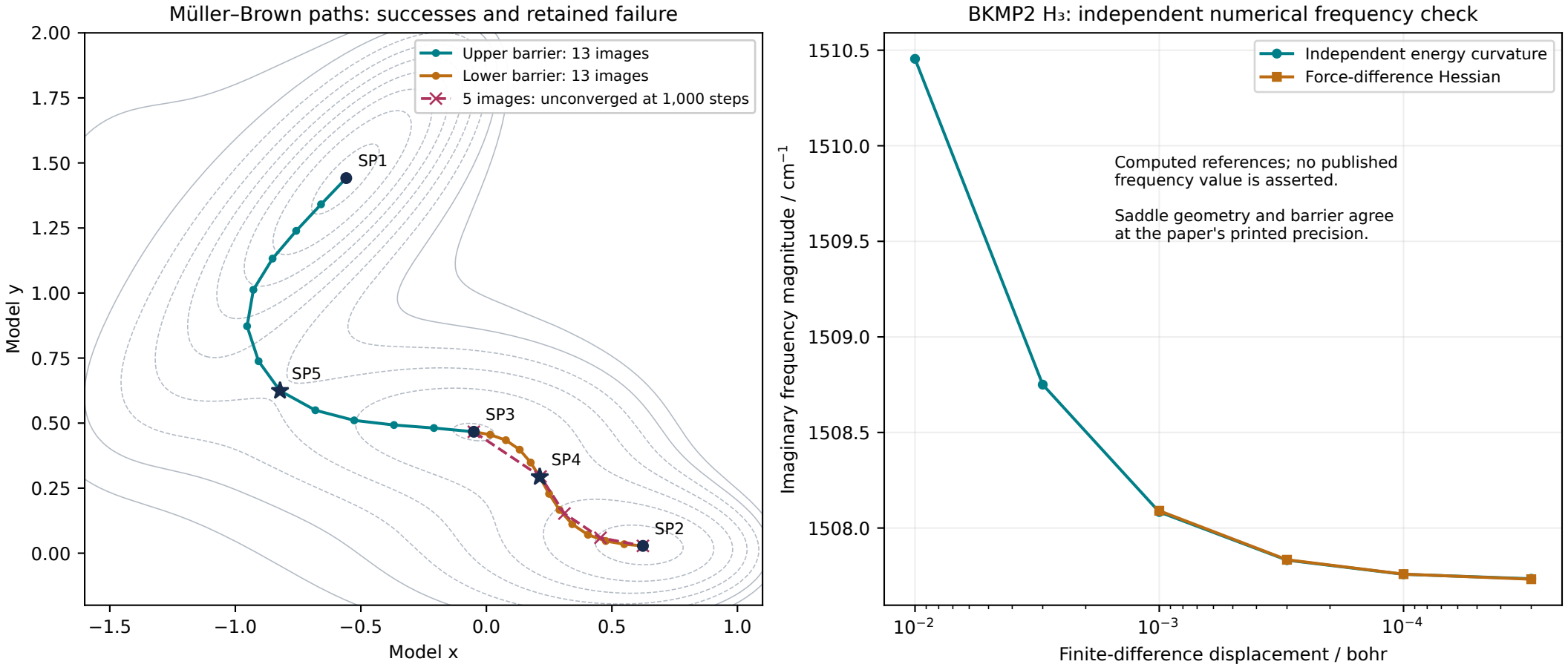
Müller–Brown paths: successes and retained failure
Upper barrier: 13 images
Lower barrier: 13 images
5 images: unconverged at 1,000 steps
SP1
SP2
SP3
SP4
SP5
Model x
Model y
BKMP2 H₃: independent numerical frequency check
Independent energy curvature
Force-difference Hessian
Computed references; no published frequency value is asserted.
Saddle geometry and barrier agree at the paper's printed precision.
Imaginary frequency magnitude / cm⁻¹
Finite-difference displacement / bohr